

Yixiao Wang

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Durham, NC 27705, United States

EDUCATION

- Duke University**
M.S. in Statistical Science
◦ GPA: 3.96/4.00
Aug. 2024 – May 2026
Durham, NC, United States
- University of Science and Technology of China, School of the Gifted Young**
B.S. in Mathematics
◦ **Top 10%** in School of the Gifted Young and the Department of Mathematics; **Outstanding Graduate**
◦ Thesis: *Trimmed Mean for Partially Observed Functional Data* (Advisor: [Prof. Xiaohong Lan](#))
Sep. 2020 – Jul. 2024
Hefei, Anhui, China

RESEARCH INTERESTS

- Interpretable and theoretically grounded machine learning**, focusing on scalable optimization and structure learning methods that unify interpretability, performance, and practicality.
- Theory-driven deep learning algorithms**, developing efficient, simple, and high-performance approaches for generative and multimodal systems, grounded in rigorous stability and approximation analyses.
- Deep learning theory**, uncovering the underlying principles and mechanisms of modern architectures to explain their generalization, reasoning, and representation behaviors from theoretical and geometric perspectives.

RESEARCH EXPERIENCE

- Duke University, Interpretable Machine Learning Lab**
Supervisor: [Prof. Cynthia Rudin](#)
Jun. 2025 – Present
Durham, NC, United States
 - Developed near-optimal algorithms for sparse regression trees with continuous features, unifying constant-leaf and linear-leaf formulations under a dynamic-programming lookahead framework. Leveraged efficient rank-one Cholesky updates to maintain ridge-regularized Gram matrix factorizations, reducing split-evaluation cost per threshold and ensuring numerical stability across large feature sets.
 - Established computational complexity bounds and investigated near-optimality guarantees for regression trees.
 - Built a complete, optimized C++/Python package with reproducible benchmarks on 10 UCI datasets, demonstrating consistent order-of-magnitude speedups over optimal baselines (OSRT, STreeD) while preserving accuracy.
- Duke University, Department of Computer Science**
Supervisor: [Prof. Anru Zhang](#)
Jun. 2025 – Present
Durham, NC, United States
 - Established the first rigorous in-context learning theory for time series forecasting, showing that Transformers are fundamentally suboptimal: proved a strictly positive excess-risk gap over linear predictors, quantified it with an inverse-linear convergence rate, and demonstrated exponential error compounding under Chain-of-Thought rollout.
 - Modeled hidden-state trajectories of Transformers using Menger curvature to analyze logical consistency, uncovering geometric invariants that govern reasoning behavior.
- Duke Center for Computational Evolutionary Intelligence (CEI)**
Supervisor: [Prof. Yiran Chen](#)
Jan. 2025 – Present
Durham, NC, United States
 - Developed **ZEUS**, a controllable training-free diffusion acceleration framework that achieves up to $3.2\times$ **end-to-end speedups** ($2.0\times\text{--}3.2\times$ across SD-2, SDXL, and Flux) while improving perceptual fidelity ($\text{LPIPS} \leq 0.08$, $\text{FID} \leq 3.5$) over prior training-free methods.
 - Developed **SADA**, a stability-guided training-free adaptive diffusion accelerator, delivering $\geq 1.8\times$ speedups across SD-2, SDXL, and Flux with $\text{LPIPS} \leq 0.10$ and $\text{FID} \leq 4.5$.
 - Contributed theoretical analysis for **FlashSVD**, deriving memory–compute complexity bounds and characterizing rank-dependent speedup and compression thresholds for low-rank Transformer inference.
- Wake Forest University, Department of Computer Science**
Supervisor: [Prof. Aditya Devarakonda](#)
Jan. 2025 – Present
Winston-Salem, NC, United States
 - Proposed and led the development of **Enhanced Cyclic Coordinate Descent (ECCD)**, a second-order Taylor-correction–based block coordinate descent algorithm for generalized linear models (GLMs) that mitigates the high computational cost from repeatedly evaluating complex link functions. Established theoretical results on computational complexity and error bounds.
 - Led the implementation of a C++-driven R package with optimized single-point and pathwise λ solvers, enabling efficient and stable regularized regression across large-scale datasets. Demonstrated superior numerical stability compared to standard block coordinate descent methods, and conducted extensive comparisons with state-of-the-art R/Python implementations.
 - Achieved consistent $2\text{--}4\times$ speedups over the widely used `glmnet` package while preserving convergence guarantees and solution accuracy.

- [C.1] Yixiao Wang*, Zishan Shao*, Ting Jiang, Aditya Devarakonda. “Enhanced Cyclic Coordinate Descent for Elastic Net GLMs,” *NeurIPS* 2025.
- [C.2] Ting Jiang*, Yixiao Wang*, Hancheng Ye*, Zishan Shao, Jingwei Sun, Jingyang Zhang, Zekai Chen, Jianyi Zhang, Yiran Chen, Hai Li. “SADA: Stability-guided Adaptive Diffusion Acceleration,” *ICML* 2025. [Code] [Page] [Slides]
- [C.3] Yufa Zhou*, Yixiao Wang*, Surbhi Goel, Anru Zhang. “Why Do Transformers Fail to Forecast Time Series In-Context?,” *NeurIPS Workshop on What Can(t) Transformers Do?* 2025 (Oral Presentation).
- [S.1] Yixiao Wang*, Hayden McTavish*, Varun Babbar*, Margo Seltzer, Cynthia Rudin. “Near-Optimal Scalable Constant and Piecewise Regression Trees,” under review at *AISTATS* 2026.
- [S.2] Yixiao Wang*, Ting Jiang*, Zishan Shao*, Hancheng Ye, Jingwei Sun, Mingyuan Ma, Jianyi Zhang, Yiran Chen, Hai Li, “ZEUS: Zero-shot Efficient Unified Sparsity for Generative Models,” under review at *ICLR* 2026. [Code] [Page]
- [S.3] Yufa Zhou*, Yixiao Wang*, Xunjian Yin*, Shuyan Zhou, Anru Zhang. “The Geometry of Reasoning: Flowing Logics in Representation Space,” under review at *ICLR* 2026. [Dataset]
- [P.1] Zishan Shao, Yixiao Wang, Qinsi Wang, Ting Jiang, Zhixu Du, Hancheng Ye, Danyang Zhuo, Yiran Chen, Hai Li. “FlashSVD: Memory-Efficient Inference with Streaming for Low-Rank Models,” under review at *AAAI* 2026. [Code]

* denotes co-first authorship; first-/co-first-authored works listed first.

HONORS AND AWARDS

- Outstanding Graduate, University of Science and Technology of China (USTC) 2024
- China Petroleum Scholarship (3 awardees, School of the Gifted Young, USTC) 2023
- Excellent Teaching Assistant (Top 3 in the University, USTC) 2023
- Silver Prize, Outstanding Student Scholarship (Top 15%, USTC) 2022 & 2021
- “Qiang Wei Feng Gong De Yu” (Diligence and Moral Conduct) Scholarship (USTC) 2022
- Excellent President of Student Club (Origami Club, USTC) 2022
- First Prize, Chinese Mathematical Competition for University Students (Top 1%) 2022 & 2021
- Bronze Prize, Outstanding Freshmen Scholarship (Top 35%, USTC) 2020

TEACHING AND OUTREACH

- **Teaching Assistant: Linear Algebra B1** Mar. 2023 – Jul. 2023
University of Science and Technology of China (USTC), Hefei, China
 - Ranked **Top 3** among all teaching assistants at USTC in the university-wide TA evaluation.
 - Built the course homepage and served as the group leader, managing a course group with over 100 students.
 - Organized weekly problem-solving sessions during weekends, delivered over 40 hours of instruction served as the group leader, and engaged with more than 200 participants, showcasing strong teaching and communication skills.
 - Independently completed solution sets for over 100 end-of-chapter exercises and provided analysis and answers to all past exam papers.
- **Origami Club President** Jun 2021 – Jun 2022
University of Science and Technology of China
 - Designed and organized the first origami exhibition “Yue Ran Zhi Shang” (Vividly Comes to Life on Paper) in USTC, responsible for venue decoration and maintaining order for two consecutive weeks.
 - Conducted origami teaching sessions twice per month, sharing the art of origami with over 200 participants.
 - Held special origami activities, such as Origami at Christmas and spray-painting completed origami works.
- **Volunteer** Sep 2021 – Jan 2022
Hefei Chunyu Parent Support Center for Intellectually Disabled Children
 - Played basketball with children with autism and helped coaches maintain order in class once per week.

SELECTED COURSE PROJECTS

- **Airbnb Price Prediction in New York City** (CS 671 Theory & Algorithms of Machine Learning, Duke University) – Achieved top 5 out of 137 participants using advanced stacking strategies, combining XGBoost, LightGBM, and 20+ weak learners for robust generalization. [Report][Code]

ADDITIONAL INFORMATION

Languages: English, Mandarin, Sichuanese dialect.

Programming & Software: Python (primary), C/C++, $\text{\LaTeX}$, Linux, HTML/CSS, MATLAB, R.

Interests: Origami and other visual arts (selected works available at [portfolio link](#)).